Neap

VCE Specialist Mathematics Units 3&4

Suggested Solutions

2025 Trial Examination 1

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Question 1 (4 marks)

Finding k by substituting into the equation gives:

$$3 = \frac{k \times 1^2}{\sqrt{4}}$$

$$3 \times 2 = k$$

$$6 = k$$ M1

The differential equation is given by $a = v\dfrac{dv}{dx}$.

$$v\frac{dv}{dx} = \frac{6x^2}{\sqrt{v}}$$

$$v^{\frac{3}{2}}\frac{dv}{dx} = 6x^2$$

$$\int v^{\frac{3}{2}}\frac{dv}{dx}dx = \int 6x^2 dx$$

$$\frac{2}{5}v^{\frac{5}{2}} = 2x^3 + c$$ M1

Using the boundary conditions to find c gives:

$$\frac{2}{5} \times 4^{\frac{5}{2}} = 2 \times 1^3 + c$$

$$\frac{2}{5} \times 32 = 2 + c$$

$$c = \frac{64}{5} - 2$$

$$= \frac{54}{5}$$ A1

Using $x = 2$ to find the object's velocity gives:

$$\frac{2}{5}v^{\frac{5}{2}} = 2 \times 2^3 + \frac{54}{5}$$

$$\frac{2}{5}v^{\frac{5}{2}} = 16 + \frac{54}{5}$$

$$\frac{2}{5}v^{\frac{5}{2}} = \frac{134}{5}$$

$$v^{\frac{5}{2}} = 67$$

$$v = 67^{\frac{2}{5}} \text{ m s}^{-1}$$ A1

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Question 2 (6 marks)

a. Using polynomial division to simplify the expression gives:

$$\begin{array}{r} q \\ x^2+5x-6\overline{\big)\,qx^2} \\ \underline{-qx^2+5qx-6q} \\ -5qx-6q \end{array}$$ M1

$$f(x)=q+\frac{-5qx-6q}{(x+6)(x-1)}$$

Therefore, the asymptotes are $y=q$, $x=-6$ and $x=1$. A1

b. Using the quotient rule to find the derivative of $f(x)$ gives:

$$f'(x)=\frac{2qx\left(x^2+5x-6\right)-(2x+5)\left(qx^2\right)}{\left(x^2+5x-6\right)^2}$$

The stationary points occur when $f'(x)=0$.

$$\begin{aligned} 0 &= \frac{2qx\left(x^2+5x-6\right)-(2x+5)\left(qx^2\right)}{\left(x^2+5x-6\right)^2} \\ &= 2qx^3+10qx^2-12qx-2qx^3-5qx^2 \\ &= 5qx^2-12qx \\ &= qx(5x-12) \end{aligned}$$

As $q\neq 0$, $x=0$ and $\frac{12}{5}$.

Therefore, the x-values of the stationary points are $x=0$ and $x=\frac{12}{5}$. A1

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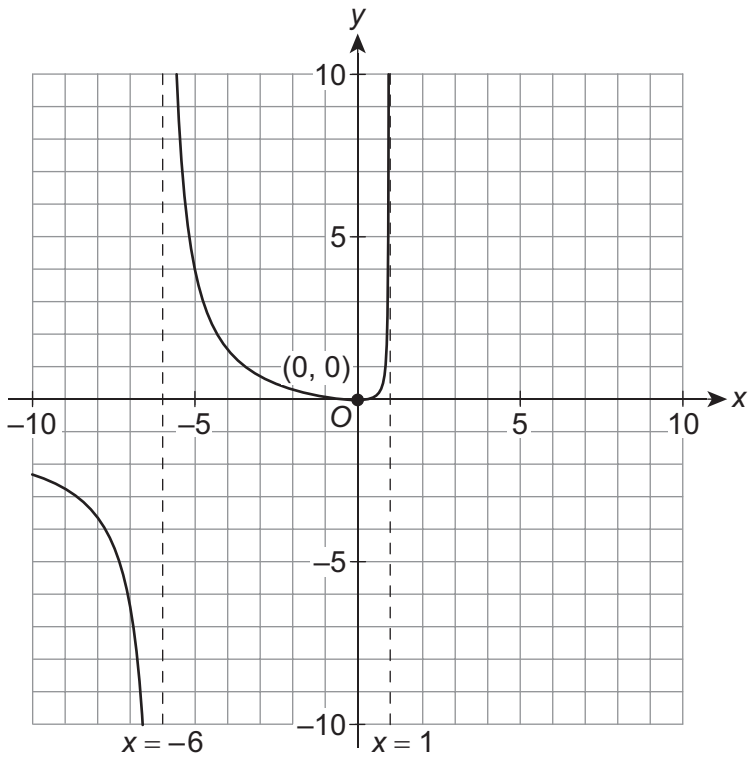
c.



correct shape A1
correct stationary point A1
correct asymptotes A1

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Question 3 (5 marks)

a. $z^3 + 64i = 0$

$(r\text{cis}(\theta))^3 = 64\text{cis}\left(-\frac{\pi}{2}\right)$

$r^3 = 64$

$r = 4$

$3\theta = -\frac{\pi}{2} + 2\pi k, k \in Z$

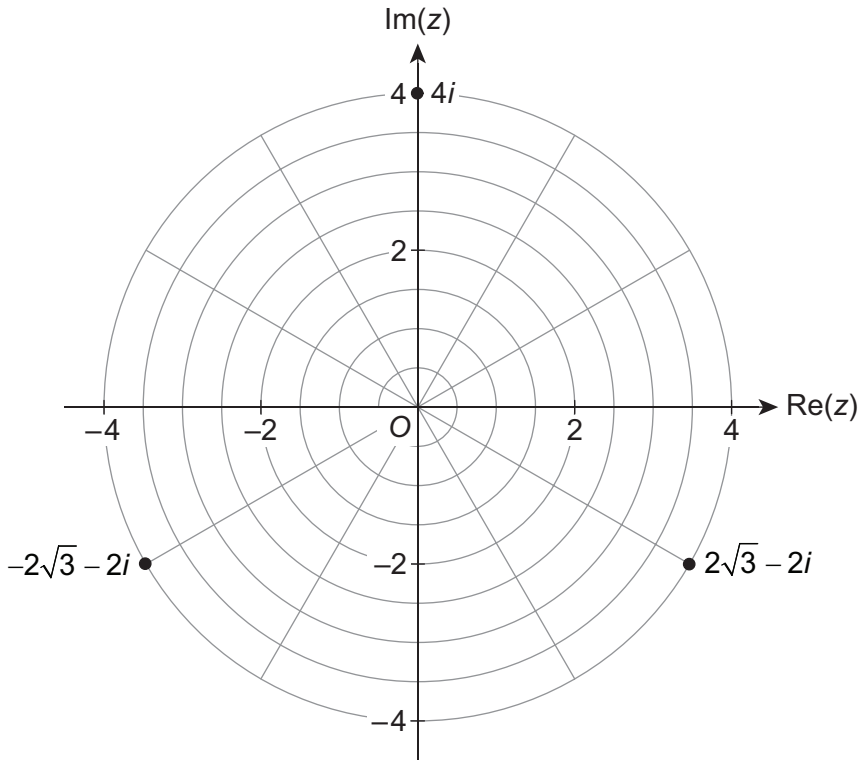
$\theta = -\frac{\pi}{6} + \frac{2\pi k}{3}$

$z = 4\text{cis}\left(-\frac{\pi}{6}\right), 4\text{cis}\left(\frac{\pi}{2}\right), 4\text{cis}\left(-\frac{5\pi}{6}\right)$ M1

$= 4\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right), 4(i), \left(\frac{-\sqrt{3}}{2} - \frac{1}{2}i\right)$

$= 2\sqrt{3} - 2i, 4i, -2\sqrt{3} - 2i$ A1

b.



A1

c. $A = \pi r^2 - \frac{r^2}{2}(\theta - \sin(\theta))$

$= \pi 4^2 - \frac{4^2}{2}\left(\frac{2\pi}{3} - \sin\left(\frac{2\pi}{3}\right)\right)$ M1

$= 16\pi - 8\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$

$= 16\pi - \frac{16\pi}{3} + 4\sqrt{3}$

$= \frac{32\pi}{3} + 4\sqrt{3}$ units^2 **OR** $\frac{32\pi + 12\sqrt{3}}{3} = \frac{4\left(8\pi + 3\sqrt{3}\right)}{3}$ units^2 A1

Question 4 (5 marks)

a. Given that $\overrightarrow{DC} = \overrightarrow{AB}$, $\overrightarrow{AB} = \underset{\sim}{i} + 2\underset{\sim}{j} - 6\underset{\sim}{k}$.

Therefore, the coordinates of C are (–2, 3, –6). A1

b. The vector normal to plane Π_1 is equal to $\overrightarrow{AB} \times \overrightarrow{AD}$.

$$\overrightarrow{AB} \times \overrightarrow{AD} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 1 & 2 & -6 \\ -2 & 1 & -3 \end{vmatrix}$$

$\underset{\sim}{n} = 15\underset{\sim}{j} + 5\underset{\sim}{k}$ A1

Therefore, plane Π_2 has the equation $15y + 5z = 0$ **OR** $3y + z = 0$. A1

c. $\overrightarrow{OD} = -3\underset{\sim}{i} + \underset{\sim}{j}$

$\underset{\sim}{n} = 3\underset{\sim}{j} + \underset{\sim}{k}$

Finding the angle between $\underset{\sim}{n}$ and $\overrightarrow{OD}$ gives:

$\cos(\theta) = \frac{\underset{\sim}{n} \cdot \overrightarrow{OD}}{|\underset{\sim}{n}||\overrightarrow{OD}|}$

$= \frac{3}{10}$ A1

Therefore, the angle between the plane and $\overrightarrow{OD}$ is $\frac{\pi}{2} - \cos^{-1}\left(\frac{3}{10}\right)$.

Finding the sine of the angle gives:

$\sin\left(\frac{\pi}{2} - \cos^{-1}\left(\frac{3}{10}\right)\right) = \cos\left(\cos^{-1}\left(\frac{3}{10}\right)\right)$

$= \frac{3}{10}$ A1

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Question 5 (4 marks)

a. L = mass of a leg, S = mass of a support, A = mass of a seat, T = total mass of a stool

Therefore:

$$T = L_1 + L_2 + L_3 + L_4 + S_1 + S_2 + A$$

$$\begin{aligned}\mathrm{E}(T) &= \mathrm{E}(L_1 + L_2 + L_3 + L_4 + S_1 + S_2 + A)\\ &= \mathrm{E}(L_1) + \mathrm{E}(L_2) + \mathrm{E}(L_3) + \mathrm{E}(L_4) + \mathrm{E}(S_1) + \mathrm{E}(S_2) + \mathrm{E}(A)\\ &= 4 \times 1200 + 2 \times 300 + 800\\ &= 4800 + 600 + 800\\ &= 6200 \text{ g}\end{aligned}$$ A1

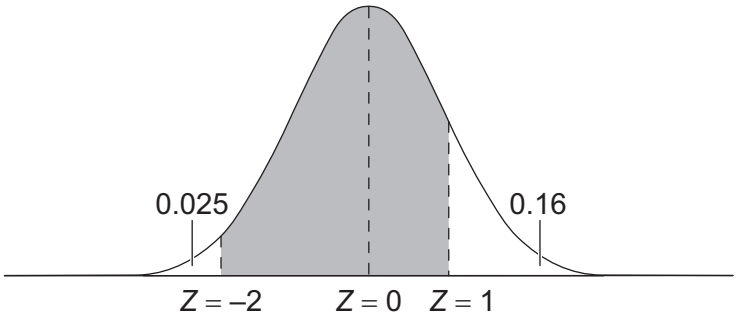
$$\begin{aligned}\mathrm{Var}(T) &= \mathrm{Var}(L_1) + \mathrm{Var}(L_2) + \mathrm{Var}(L_3) + \mathrm{Var}(L_4) + \mathrm{Var}(S_1) + \mathrm{Var}(S_2) + \mathrm{Var}(A)\\ &= 4 \times 4^2 + 2 \times 2^2 + 3^2\\ &= 81\end{aligned}$$

$$\begin{aligned}\sigma(T) &= \sqrt{\mathrm{Var}(T)}\\ &= 9 \text{ g}\end{aligned}$$ A1

b. $$\begin{aligned}\bar{X} &\sim N\left(6200, \frac{9}{\sqrt{36}}\right)\\ &\sim N(6200, 1.5)\end{aligned}$$

$$\begin{aligned}\Pr(6197 < \bar{X} < 6201.5) &= \Pr\left(\frac{6197 - 6200}{1.5} < Z < \frac{6201.5 - 6200}{1.5}\right)\\ &= \Pr(-2 < Z < 1)\end{aligned}$$ M1

Sketching the probability distribution gives:



$$\begin{aligned}\Pr(6197 < \bar{X} < 6201.5) &= \text{shaded area}\\ &= 1 - 0.025 - 0.16\\ &= 0.815\end{aligned}$$ A1

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Question 6 (3 marks)

Let $P(n)$ be the proposition $1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$.

Proving $P(n)$ is true for $n = 1$ gives:

$$\begin{aligned}\text{LHS} &= 1^3\\ &= 1\end{aligned}$$

$$\begin{aligned}\text{RHS} &= \left(\frac{1(1+1)}{2}\right)^2\\ &= \left(\frac{1\times 2}{2}\right)^2\\ &= 1^2\\ &= 1\end{aligned}$$

LHS = RHS; therefore, $P(1)$ is true. A1

Assuming $P(n)$ is true for $n = k$ gives:

$$1^3 + 2^3 + 3^3 + \cdots + k^3 = \left(\frac{k(k+1)}{2}\right)^2,\ k \in N$$

Proving $P(n)$ is true for $n = k + 1$ gives:

$$1^3 + 2^3 + 3^3 + \cdots + k^3 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2}\right)^2,\ k \in N$$

$$\begin{aligned}\text{LHS} &= 1^3 + 2^3 + 3^3 + \cdots + k^3 + (k+1)^3\\ &= \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 && \text{M1}\\ &= (k+1)^2\left(\left(\frac{k}{2}\right)^2 + (k+1)\right)\\ &= (k+1)^2\left(\frac{k^2}{4} + k + 1\right)\\ &= \frac{1}{4}(k+1)^2\left(k^2 + 4k + 4\right)\\ &= \frac{1}{4}(k+1)^2(k+2)^2\\ &= \frac{(k+1)^2(k+2)^2}{2^2}\\ &= \left(\frac{(k+1)(k+2)}{2}\right)^2\\ &= \text{RHS} && \text{A1}\end{aligned}$$

If k is true, then $k + 1$ is true.

Therefore, by mathematical induction, $1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$ is true for all $n \in N$.

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Question 7 (3 marks)

$$\sin\left(e^y\right)+x\times\cos\left(e^y\right)\times e^y\times\frac{dy}{dx}+3x^2\log_e\left(\frac{1}{y}\right)+x^3\times\frac{1}{\frac{1}{y}}\times -y^{-2}\times\frac{dy}{dx}=0$$ M1

$$\frac{dy}{dx}\left(xe^y\cos\left(e^y\right)-\frac{x^3}{y}\right)=-\sin\left(e^y\right)-3x^2\log_e\left(\frac{1}{y}\right)$$

$$\frac{dy}{dx}=\frac{3x^2\log_e(y)-\sin\left(e^y\right)}{xe^y\cos\left(e^y\right)-\frac{x^3}{y}}$$ M1

$$=\frac{y\left(3x^2\log_e(y)-\sin\left(e^y\right)\right)}{xye^y\cos\left(e^y\right)-x^3}$$ A1

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Question 8 (3 marks)

$$S = 2\pi\int_2^4 x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$

$$= 2\pi\int_2^4 6t\sqrt{6^2 + \left(2t - \frac{9}{2t}\right)^2}\,dt \qquad \text{M1}$$

$$= 12\pi\int_2^4 t\sqrt{36 + 4t^2 - 18 + \frac{81}{4t^2}}\,dt$$

$$= 12\pi\int_2^4 t\sqrt{4t^2 + 18 + \frac{81}{4t^2}}\,dt$$

$$= 12\pi\int_2^4 t\sqrt{\left(2t + \frac{9}{2t}\right)^2}\,dt$$

$$= 12\pi\int_2^4 t\left(2t + \frac{9}{2t}\right)dt$$

$$= 12\pi\int_2^4 2t^2 + \frac{9}{2}\,dt$$

$$= 12\pi\left[\frac{2t^3}{3} + \frac{9t}{2}\right]_2^4 \qquad \text{M1}$$

$$= 12\pi\left[\frac{2\times 4^3}{3} + \frac{9\times 4}{2} - \frac{2\times 2^3}{3} - \frac{9\times 2}{2}\right]$$

$$= 12\pi\left[\frac{128}{3} + \frac{36}{2} - \frac{16}{3} - \frac{18}{2}\right]$$

$$= 12\pi\left[\frac{112}{3} + 9\right]$$

$$= 12\pi\left[\frac{112}{3} + \frac{27}{3}\right]$$

$$= 12\pi\left[\frac{139}{3}\right]$$

$$= 4\pi \times 139$$

$$= 556\pi \text{ units}^2 \qquad \text{A1}$$

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Question 9 (4 marks)

$$\int_{-1}^{\sqrt{3}-2} \frac{x}{x^2+4x+5}dx$$

$$= \frac{1}{2}\int_{-1}^{\sqrt{3}-2} \frac{2x+4}{x^2+4x+5}dx - \int_{-1}^{\sqrt{3}-2} \frac{2}{x^2+4x+5}dx \qquad \text{M1}$$

$$= \frac{1}{2}\left[\log_e\left|x^2+4x+5\right|\right]_{-1}^{\sqrt{3}-2} - 2\int_{-1}^{\sqrt{3}-2} \frac{1}{(x+2)^2+1}dx$$

$$= \frac{1}{2}\left[\log_e\left|\left(\sqrt{3}-2\right)^2+4\left(\sqrt{3}-2\right)+5\right| - \log_e\left|(-1)^2+4(-1)+5\right|\right] - 2\left[\tan^{-1}(x+2)\right]_{-1}^{\sqrt{3}-2} \qquad \text{M1}$$

$$= \frac{1}{2}\left[\log_e\left|3-4\sqrt{3}+4+4\sqrt{3}-8+5\right| - \log_e\left|1-4+5\right|\right] - 2\left[\tan^{-1}\left(\sqrt{3}-2+2\right) - \tan^{-1}(-1+2)\right]$$

$$= \frac{1}{2}\left[\log_e|4| - \log_e|2|\right] - 2\left[\tan^{-1}\left(\sqrt{3}\right) - \tan^{-1}(1)\right]$$

$$= \frac{1}{2}\left[\log_e(2)\right] - 2\left[\frac{\pi}{3} - \frac{\pi}{4}\right] \qquad \text{M1}$$

$$= \log_e\left(\sqrt{2}\right) - 2\left(\frac{4\pi}{12} - \frac{3\pi}{12}\right)$$

$$= \log_e\left(\sqrt{2}\right) - 2\left(\frac{\pi}{12}\right)$$

$$= \log_e\left(\sqrt{2}\right) - \frac{\pi}{6} \qquad \text{A1}$$

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Question 10 (3 marks)

a. $x = 2\cos(2t)$

$$\frac{x}{2} = \cos(2t)$$

$$y = \sin^2(t) - 1$$

$$y = \frac{1}{2}(1 - \cos(2t)) - 1$$

$$y = \frac{1}{2}\left(1 - \frac{x}{2}\right) - 1$$

$$y = \frac{1}{2} - \frac{x}{4} - 1$$

$$4y = 2 - x - 4$$

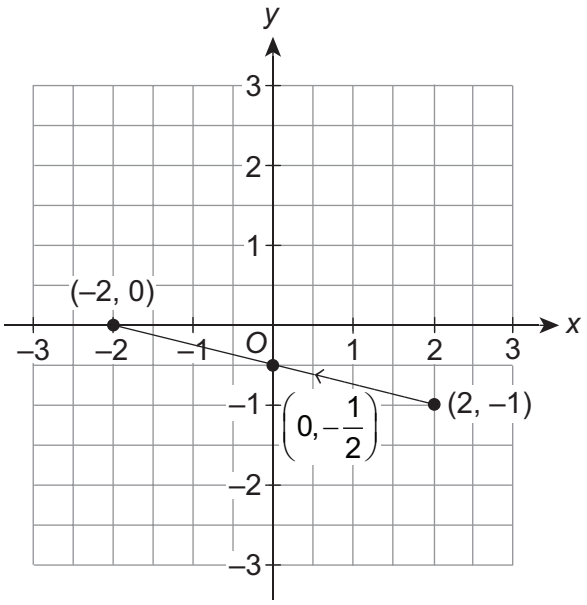
$$x + 4y = -2$$ A1

b. The range of the relation is $x = 2\cos(2t)$; therefore, the domain of the relation is $x = 2\cos(2t)$, where $t \in \left[0, \frac{\pi}{2}\right]$ and $x \in [0, 2]$.

At $t = 0$, $x = 2$ and $y = -1$.

At $t = \frac{\pi}{2}$, $x = 0$ and $y = -\frac{1}{2}$.

Therefore:



A1

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c. $\dot{\underset{\sim}{r}}(t) = -4\sin(2t)\underset{\sim}{i} + 2\sin(t)\cos(t)\underset{\sim}{j}$

$$\left|\dot{\underset{\sim}{r}}(t)\right| = \sqrt{\left(-4\sin(2t)\right)^2 + \left(\sin(2t)\right)^2}$$

$$= \sqrt{16\sin^2(2t) + \sin^2(2t)}$$

$$= \sqrt{17\sin^2(2t)}$$

$$= \sqrt{17}\sin(2t)$$

$$\sqrt{17}\sin(2t) = \frac{\sqrt{17}}{2}$$

$$\sin(2t) = \frac{1}{2}$$

$$2t = \frac{\pi}{6}$$

$$t = \frac{\pi}{12}\text{ s}$$ A1